Notes: Pohl, Schmedders, and Wilms (RFS, 2024)

Existence of the Wealth-Consumption Ratio in Asset Pricing Models with Recursive Preferences

Contents

1 Big picture

- **Question.** In modern asset-pricing models with recursive preferences, when is lifetime utility actually well defined? Equivalently, when does the wealth-consumption ratio stay finite rather than blow up?
- **Why this matters.** A huge share of quantitative asset-pricing uses Epstein–Zin or related recursive preferences together with persistent growth, stochastic volatility, jumps, disasters, learning, or ambiguity. If the utility recursion has no finite solution, the model is not just hard to solve - it is conceptually ill posed.
- **Core challenge.** Existence is delicate. The paper's central warning is that tiny-looking changes in the consumption process can destroy existence. That is why numerical solution methods alone can be misleading: they may return an object even when the underlying infinite-horizon model has no well-defined solution.
- **Main contribution.** The paper develops an order-theoretic method that delivers *existence*, *nonexistence*, and *uniqueness* results for recursive utility. The approach is based on two comparison tools:
  - a **Minkowski bound**, which uses superadditivity or subadditivity of the recursive operator to construct a lower or upper bound,
  - a **Jensen bound**, which compares the recursive utility problem to a simpler benchmark, typically CRRA utility.
- **Main theoretical result.** The paper shows how to sandwich the wealth-consumption ratio between two objects that are analytically tractable. In the affine class, these bounds can be written in closed form, so the existence question becomes a clean condition on preference parameters and consumption-process parameters.
- **Uniqueness result.** The relevant solution is not merely "some" fixed point. It is the *limit of finite-horizon solutions*, the *smallest nonnegative solution*, and the unique homogeneous value function that does not embed a rational bubble.
- **Key practical message.** For the parameterizations used in leading papers — Bansal and Yaron, Drechsler and Yaron, Wachter, Ju and Miao, and Cecchetti-Lam-Mark — the paper can usually settle the existence question directly.
- **Main applied findings.**
  - Standard Bansal-Yaron long-run-risk calibrations have a well-defined solution.
  - Adding stochastic-volatility shocks or volatility jumps does not generally break existence for the benchmark calibrations used in practice.
  - Wachter-style time-varying disaster models also satisfy the existence condition for empirically relevant parameter values.
  - Ju-Miao smooth-ambiguity learning models converge under the benchmark calibration.
  - In contrast, the Collin-Dufresne, Johannes, and Lochstoer learning model with an unbounded conjugate normal prior and $\psi \neq 1$ diverges.

- **What is intellectually distinctive.** The paper does *not* rely on contraction mappings on bounded state spaces. Instead, it exploits homogeneity of utility and order properties of the recursive problem. This allows the analysis to handle unbounded Gaussian state spaces and non-Gaussian shocks such as jumps.
- **Main economic lesson.** The discount factor, the EIS, the mean growth rate, and each source of persistent risk all push the wealth-consumption ratio in interpretable directions. The paper turns "does the model exist?" from a vague numerical concern into a comparative-statics question.
- **Takeaway.** Before trusting a recursive-preferences asset-pricing model, check existence. This paper gives a practical toolkit for doing that and shows that many influential calibrations do survive the test, though some do not.

2 Data and measurement (key objects)

This is a theory paper, so the "data" are not micro observations or macro panels in the usual sense. The key objects are recursive utility operators, transformed value functions, and parameterized consumption processes.

2.1 The generic recursive object

The paper starts from a very general recursion

$$v_t = S_t v_{t+1}, \tag{1}$$

where S_t maps a time-$(t+1)$ random variable into a time-t random variable.

Interpretation. The object v_t is the continuation value in normalized form. In most applications it becomes the wealth-consumption ratio or a closely related transform of lifetime utility.

2.2 Finite-horizon solutions and the economically relevant limit

For a terminal horizon T, define

$$v_{t,T} = S_t v_{t+1,T} \quad \text{for } t < T, \qquad v_{T,T} = S_T 0. \tag{2}$$

The infinite-horizon solution of interest is then

$$v_t = \lim_{T\to\infty} v_{t,T}. \tag{3}$$

Interpretation. This is crucial. The paper does not regard every formal fixed point as equally meaningful. The preferred solution is the one obtained as the limit of finite-horizon economies.

2.3 The Monotone Convergence Property (MCP)

The one technical property repeatedly used is the **Monotone Convergence Property**. Informally, the operator S_t must preserve limits of monotone sequences of positive random variables.

Expectations and the standard recursive utility aggregators built from expectations satisfy MCP.

Interpretation. MCP is what lets the authors pass from finite-horizon bounds to infinite-horizon bounds.

2.4 Epstein–Zin parameters

For Epstein–Zin preferences the paper uses the standard parametrization

$$\alpha = 1 - \frac{1}{\psi}, \qquad \theta = \frac{1-\gamma}{1-1/\psi}, \tag{4}$$

where:

- ψ is the elasticity of intertemporal substitution (EIS),
- γ is relative risk aversion,
- δ is the time discount factor.

The paper pays special attention to the empirically important case

$$\psi > 1, \qquad \gamma > 1, \qquad \gamma > \frac{1}{\psi}, \tag{5}$$

which implies $\theta < 0$ and corresponds to a preference for early resolution of risk.

Interpretation. This is exactly the parameter region used in much of the long-run-risks literature, so the paper is speaking directly to the leading quantitative models.

2.5 Normalized wealth-consumption ratio

Under Epstein–Zin, the normalized object is

$$v_t = \frac{V_t^\alpha}{(1-\delta)C_t^\alpha}, \tag{6}$$

where V_t is lifetime utility and C_t is consumption.

Interpretation. This normalization converts the utility recursion into an equation for the wealth-consumption ratio, which is the natural object for proving finiteness.

2.6 Affine state processes

A major part of the paper is built for discrete-time affine processes. The state vector is written as

$$X_t = (X_{0,t}, X_{1,t}, X_{2,t}, \ldots), \qquad X_{0,t} \equiv 1, \tag{7}$$

with $X_{1,t}$ often identified with consumption growth and the remaining states capturing long-run mean growth, stochastic volatility, disaster intensity, or other latent drivers.

The defining affine property is

$$E_t\left[e^{k \cdot X_{t+1}}\right] = e^{h(k) \cdot X_t} \tag{8}$$

for some function $h(\cdot)$.

Interpretation. This is what makes the bounds analytically tractable. Once the operator acts on exponential-linear functions and returns exponential-linear functions, the infinite-horizon bound becomes a recursion for coefficient vectors.

2.7 The key convergence statistic

For affine models the existence check is summarized by a statistic

$$\Omega(\theta) \equiv \lim_{i \to \infty} \sup (A_{i+1}(\theta) - A_i(\theta)) \cdot X_t. \tag{9}$$

Roughly speaking:

- if the relevant Ω is negative, the bound converges,
- if the relevant Ω is positive, the bound diverges.

Interpretation. The entire existence question is reduced to the sign of a limit that can be read directly from the model's parameters.

2.8 The benchmark models the paper evaluates

The paper applies the framework to five main classes of models:

- long-run risk with stochastic volatility and jumps,
- time-varying consumption-disaster models,
- Markov-switching learning models,
- smooth ambiguity-aversion models,
- random-walk learning models with unbounded priors.

Interpretation. The paper is not merely abstract. It is designed to say whether the influential quantitative models used in the literature actually exist.

3 Models and empirical specifications (all equations + interpretation)

3.1 The abstract recursion

The generic recursive equation is

$$v_t = S_t v_{t+1}. \tag{1}$$

A solution means that the entire sequence $(v_t)_{t \geq 0}$ is well defined and finite.

The limit solution is obtained from finite horizons:

$$v_t = \lim_{T \to \infty} v_{t,T}. \tag{2}$$

Interpretation. The paper treats the infinite-horizon problem as the limit of a sequence of finite-horizon problems, rather than as a standalone fixed-point problem.

3.2 The Minkowski bound

Suppose the operator can be decomposed as

$$S_t x_{t+1} = G_t + H_t x_{t+1}, \tag{10}$$

where H_t is either *subadditive* or *superadditive*:

$$H_t(x_{t+1} + y_{t+1}) \leq H_t x_{t+1} + H_t y_{t+1}, \tag{3}$$
$$H_t(x_{t+1} + y_{t+1}) \geq H_t x_{t+1} + H_t y_{t+1}. \tag{4}$$

Define

$$g_{T,T} = G_T, \qquad g_{t,T} = H_t g_{t+1,T}, \tag{11}$$

and

$$w_t = \sum_{i \in \mathbb{N}_0} g_{t,t+i}. \tag{5}$$

Then:

- if H_t is subadditive, the limit solution satisfies $v_t \leq w_t$,
- if H_t is superadditive, the limit solution satisfies $v_t \geq w_t$,
- if H_t is additive, then $v_t = w_t$.

Interpretation. This is the first main theorem. It gives either an upper bound or a lower bound, depending on the curvature of the recursion. Finite upper bounds prove existence; infinite lower bounds prove nonexistence.

3.3 The Jensen bound

The second main comparison result considers two operators S_t and R_t satisfying

$$0 \leq S_t x_{t+1} \leq S_t y_{t+1} \leq R_t y_{t+1} \quad \text{for } 0 \leq x_{t+1} \leq y_{t+1}. \tag{6}$$

If the benchmark recursion under R_t has a nonnegative solution, then so does the original recursion under S_t.

Interpretation. This is the Jensen bound. In practice, the benchmark is usually CRRA utility. So if the CRRA model exists, that can be enough to prove existence for a more complicated recursive utility model.

3.4 Uniqueness through the Bellman equation

For homogeneous recursive utility,

$$U_t = \frac{C_t^\alpha}{\alpha} + H_t U_{t+1}, \tag{7}$$

which is rewritten as

$$S_t x_{t+1} = \frac{C_t^\alpha}{\alpha} + H_t x_{t+1}. \tag{8}$$

The associated portfolio problem has Bellman equation

$$J_t(W_t) = \max_{C_t, W_{t+1}} \left\{ \frac{C_t^\alpha}{\alpha} + H_t J_{t+1}(W_{t+1}) \right\}. \tag{9}$$

Theorem 3 states that the economically relevant solution is homogeneous of degree α in wealth and is the smallest positive solution (if $\alpha > 0$) or the largest negative solution (if $\alpha < 0$). Any other finite solution implies a bubble in the optimal wealth path.

Interpretation. This is how the paper closes the uniqueness problem. The no-bubble value function picks out one solution from the potentially many mathematical fixed points.

3.5 Specialization to Epstein–Zin preferences

Epstein–Zin preferences are written as

$$V_t = \left[(1-\delta) C_t^\alpha + \delta \left(E_t\left[V_{t+1}^{\alpha\theta} \right] \right)^{1/\theta} \right]^{1/\alpha}. \tag{10}$$

After normalization,

$$v_t = \frac{V_t^\alpha}{(1-\delta) C_t^\alpha}, \tag{11}$$

so the wealth-consumption ratio satisfies

$$v_t = 1 + \delta \left(E_t\left[e^{\alpha\theta \Delta c_{t+1}} v_{t+1}^\theta \right] \right)^{1/\theta}. \tag{12}$$

The corresponding operators are

$$S_t x_{t+1} = 1 + \delta \left(E_t\left[e^{\alpha\theta \Delta c_{t+1}} x_{t+1}^\theta \right] \right)^{1/\theta}, \tag{13}$$

$$H_t x_{t+1} = \delta \left(E_t\left[e^{\alpha\theta \Delta c_{t+1}} x_{t+1}^\theta \right] \right)^{1/\theta}. \tag{14}$$

For Epstein–Zin, Minkowski's inequality implies:

- if $\theta \geq 1$, then H_t is subadditive and the Minkowski bound is an upper bound,
- if $\theta \leq 1$, then H_t is superadditive and the Minkowski bound is a lower bound.

The CRRA benchmark is

$$R_t x_{t+1} = 1 + \delta E_t\left[e^{\alpha \Delta c_{t+1}} x_{t+1} \right]. \tag{15}$$

Corollary 2 then says:

- if CRRA exists, then Epstein–Zin exists for $\theta < 1$,
- if CRRA does not exist, then Epstein–Zin cannot exist for $\theta > 1$.

Interpretation. In the quantitatively important case $\theta < 0$, CRRA finiteness is enough to prove existence. That is why the CRRA comparison becomes so powerful.

3.6 Affine processes and closed-form bounds

For affine state dynamics,

$$E_t\left[e^{k\cdot X_{t+1}}\right] = e^{h(k)\cdot X_t}. \tag{12}$$

Let $\tau = [\ln\delta, \alpha, 0, \ldots]$. Then for Epstein–Zin,

$$H_t e^{k\cdot X_{t+1}} = e^{h(\theta(\tau+k))/\theta\cdot X_t}. \tag{16}$$

Hence the terms in the Minkowski sum take the form

$$g_i(X_t) = \exp\{A_i(\theta)\cdot X_t\}, \tag{13}$$

with recursion

$$A_0(\theta) = 0, \qquad A_{i+1}(\theta) = \frac{h((A_i(\theta)+\tau)\theta)}{\theta}. \tag{14}$$

The bound is therefore

$$w_t(\theta) = \sum_{i=0}^{\infty} e^{A_i(\theta)\cdot X_t}. \tag{17}$$

The convergence test is summarized by

$$\Omega(\theta) = \limsup_{i\to\infty}(A_{i+1}(\theta) - A_i(\theta))\cdot X_t. \tag{18}$$

Theorem 5 gives the operational rule:

- for $\theta < 1$, if $\Omega(1) < 0$ then a solution exists, and if $\Omega(\theta) > 0$ then no solution exists;
- for $\theta > 1$, the roles reverse: $\Omega(\theta) < 0$ proves existence and $\Omega(1) > 0$ proves nonexistence.

Interpretation. This is the paper's practical engine. Once the model is affine, checking existence is reduced to the sign of one limit.

3.7 Long-run risk with stochastic volatility

The baseline long-run-risk environment is

$$\begin{aligned}
\Delta c_{t+1} &= \mu + x_t + \sigma_t\varepsilon_{c,t+1},\\
x_{t+1} &= \rho x_t + \phi_x\sigma_t\varepsilon_{x,t+1},\\
\sigma^2_{t+1} &= \bar{\sigma}^2(1-\rho_\sigma) + \rho_\sigma\sigma^2_t + \varepsilon_{\sigma,t+1},
\end{aligned} \tag{20}$$

with different admissible distributions for the volatility shock $\varepsilon_{\sigma,t+1}$.

Proposition 2 shows that

$$\Omega(\theta) = \ln\delta + \alpha\mu + \frac{1}{2}\theta\alpha^2\bar{\sigma}^2 + \frac{1}{2}\theta\alpha^2\bar{\sigma}^2\frac{\phi_x^2}{(1-\rho)^2} + \frac{1}{\theta}\ln M_{\varepsilon_\sigma}(\cdot), \tag{15}$$

where the last term captures the incremental effect of stochastic volatility through the moment-generating function of the volatility shock.

Interpretation. The first four terms are familiar: discounting, mean growth, ordinary consumption risk, and persistent long-run growth risk. The new term isolates the existence effect of stochastic volatility.

3.8 Time-varying disaster models

For Wachter-style disasters,

$$\begin{aligned} \Delta c_{t+1} &= \mu + \sigma\varepsilon_{t+1} + J_{t+1}, \\ d\lambda_t &= \kappa(\bar{\lambda} - \lambda_t)dt + \phi_\ell\sqrt{\lambda_t}dB_t, \end{aligned} \tag{21}$$

where J_{t+1} is the consumption-disaster jump and λ_t is the time-varying disaster intensity.

Proposition 3 gives a closed-form expression for $\Omega(\theta)$ consisting of a discounting term, mean-growth term, Gaussian risk term, and a disaster-jump term driven by the square-root intensity process.

Interpretation. Time-varying disaster risk changes existence mainly by shifting the effective growth term and the jump correction. For the benchmark calibration, however, the model still exists.

3.9 Learning and smooth ambiguity aversion

The Ju-Miao smooth-ambiguity utility is

$$V_t = \left[(1-\delta)C_t^{1-1/\psi} + \delta\left(E\left(\left[E\left(V_{t+1}^{1-\gamma} \mid s_t, \pi\right)\right]^{\frac{1-\eta}{1-\gamma}} \,\Big|\, s_t\right)\right)^{\frac{1-1/\psi}{1-\eta}}\right]^{\frac{1}{1-1/\psi}}. \tag{22}$$

Here η is the ambiguity-aversion parameter. The paper shows that the same Jensen and Minkowski logic can be extended to this class by applying Jensen and Minkowski inequalities twice.

For the two-state Markov-switching learning model of Ju and Miao, consumption growth follows

$$\Delta c_{t+1} = \mu_{z_{t+1}} + \sigma\varepsilon_{t+1}. \tag{27}$$

Proposition 5 yields the simple existence condition

$$\Omega(1) = \ln\delta + B + \frac{1}{2}\alpha^2\sigma^2 < 0, \tag{16}$$

where B is a weighted average growth term built from the transition matrix and the two state-specific means.

Interpretation. The regime-switching model behaves like a growth model with an effective mean growth term somewhere between the good-state and bad-state means.

3.10 Learning with an unbounded prior

For the Collin-Dufresne, Johannes, and Lochstoer setup, consumption growth is i.i.d. normal conditional on an unknown mean μ, and the investor has a conjugate normal prior over μ. In that case the terms in the Minkowski bound take the form

$$g_{t,T} = \delta^{T-t} e^{\frac{1}{2}\alpha^2\theta\sigma^2(T-t)} M(\alpha\theta(T-t))^{1/\theta}, \tag{17}$$

where $M(\cdot)$ is the moment-generating function of the prior.

Under the normal prior, $M(x)$ grows too fast. The log ratio of consecutive terms tends to infinity, so the infinite sum diverges. The paper therefore proves that with an unbounded conjugate normal prior and $\psi \neq 1$, the model has no finite solution. Under smooth ambiguity aversion, the divergence result remains.

Interpretation. This is one of the paper's sharpest applications: learning plus an unbounded prior can make asset prices explode even if each individual candidate model looks harmless on its own.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no IV in the usual sense

This paper is not an empirical identification paper. There is no instrument, no quasi-experiment, and no causal panel design. Credibility comes from exact logical implications of the recursive structure.

Interpretation. "Identification" here means identifying the economically relevant fixed point and the sign of the bounds, not recovering a treatment effect.

4.2 Why the bounds are credible

The credibility of the existence results rests on three transparent ingredients:

1. the recursive operator satisfies MCP,
2. curvature of the aggregator determines whether the Minkowski construction is an upper or lower bound,
3. CRRA provides a benchmark recursion for Jensen-type comparison.

None of these steps depends on numerical approximation error.

Interpretation. The paper's main gain over "solve it numerically and hope" is that the logic is analytical and monotone.

4.3 Why the finite-horizon limit matters for uniqueness

The recursion can have multiple solutions. Some are economically pathological:

- nonhomogeneous solutions that break the tight link between wealth and utility,
- homogeneous but exotic solutions that effectively value asymptotic properties at infinity,
- solutions associated with bubbles in the optimal wealth portfolio.

The paper rules these out by requiring that the value function be the limit of finite-horizon problems and that the implied optimal portfolio satisfy a no-bubble condition.

Interpretation. This is why the uniqueness result is meaningful economically rather than merely formal mathematically.

4.4 Why affine structure is so important

The whole method becomes operational once the model is affine. Affinity means:

- the Minkowski terms remain exponential-linear,
- the infinite-horizon problem collapses to a coefficient recursion,
- the Cauchy ratio test turns convergence into the sign of $\Omega(\theta)$.

Interpretation. The paper is strongest precisely because most quantitative macro-finance models already live in the affine class.

4.5 Why the paper improves on earlier approaches

The paper positions itself against two limitations of existing methods:

- contraction-mapping approaches often require restrictive metrics or bounded state spaces,
- spectral-radius approaches can require bounded state spaces and often need numerical computation of the condition itself.

By contrast, the present method:

- handles unbounded Gaussian state spaces,
- handles non-Gaussian shocks such as jumps,
- delivers existence and nonexistence in interpretable closed form.

Interpretation. The paper trades a necessary-and-sufficient condition for a pair of sharp, easy-to-check sufficient conditions that work in the models people actually use.

4.6 How tight are the bounds?

A reasonable concern is that upper and lower bounds might be too far apart to be useful. The paper answers this directly by studying the long-run-risk example and Figure 1. For empirically relevant values, the black "inconclusive" region is small. Standard calibrations lie comfortably in the existence region.

Interpretation. The method is not only elegant; it is practically informative.

4.7 What the paper can and cannot do

The paper can:

- prove existence,
- prove nonexistence,
- establish uniqueness of the no-bubble homogeneous solution,
- compare how discounting, growth, persistence, volatility, jumps, and ambiguity affect existence.

The paper cannot:

- say much about production economies without additional bounds on the endogenous consumption process,
- guarantee that every unmodeled numerical algorithm will find the correct solution,
- replace the need to think carefully about parameterization of the consumption process.

Interpretation. The paper solves an important piece of the recursive-utility problem, but it is still an endowment-economy toolkit.

4.8 Relation to the literature

Relative to Epstein and Zin (1989), Marinacci and Montrucchio (2010), Borovicka and Stachurski, Christensen (2022), and the order-theoretic literature, this paper's distinctive contribution is to combine:

- the homogeneity-based wealth-consumption transformation,
- explicit upper and lower bounds,
- analytical affine-model formulas,
- and a no-bubble uniqueness refinement.

Interpretation. It is less about inventing recursive utility from scratch and more about making today's recursive-preferences asset-pricing models mathematically checkable.

5 Tables and figures

This paper is mostly **theorem- and proposition-driven**. There are no conventional empirical tables. The visual evidence is concentrated in **Figure 1**, while the quantitative content is organized around explicit propositions and worked calibrations.

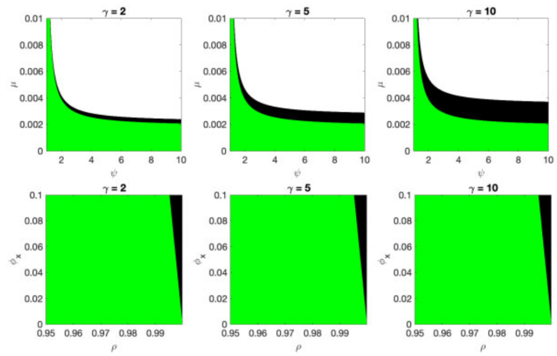

Figure 1
Existence with long-run consumption risks
The figure plots regions of existence (green) and nonexistence (white) and regions in which no definite answer on existence can be given (black) for model (19). The model parameters are given by $\delta = 0.998, \psi = 1.5, \mu = 0.0015, \bar{\sigma} = 0.0078, \rho_x = 0.979, \phi_x = 0.044$, and $l_0 = 0$.

5.1 Figure 1: existence regions in long-run-risk models

Read:

- Figure 1 plots three parameter slices for $\gamma \in \{2, 5, 10\}$.
- The green region is where $\Omega(1) < 0$, so existence is guaranteed.
- The white region is where $\Omega(\theta) > 0$, so nonexistence is guaranteed.
- The black region is where the two sufficient conditions do not settle the question.
- For realistic calibrations the black region is small, especially when the mean growth rate is not implausibly high.

Economic message. The two bounds are tight enough to be useful. For the calibrations used in practice, existence is usually not an ambiguous matter.

5.2 Proposition 2: long-run risk with stochastic volatility

Read:

- Proposition 2 decomposes $\Omega(\theta)$ into discounting, mean-growth, consumption-risk, persistent-growth, and volatility-shock terms.
- The volatility term depends on the moment-generating function of the volatility shock.
- The sign logic is intuitive: more discounting slackens existence, stronger growth helps existence, while persistent risk terms generally tighten it.

Economic message. The proposition turns a complicated recursive model into a comparative-statics formula. You can literally read from the equation how each structural feature shifts the existence margin.

5.3 Bansal-Yaron application and the three volatility cases

Read:

- For the benchmark Bansal-Yaron calibration, the paper shows that a solution exists.
- This conclusion is robust across three ways of forcing volatility nonnegativity: gamma shocks, truncated normal shocks, and Poisson jumps.
- Truncating normal volatility shocks has a negligible quantitative effect on the existence margin for the baseline calibration.
- Volatility jumps matter only modestly for existence, even under very persistent volatility.

Economic message. The paper is not rescuing the benchmark by special distributional choices. The standard long-run-risk calibration is genuinely on the "exists" side.

5.4 Proposition 3: time-varying disaster intensity

Read:

- Proposition 3 derives the existence statistic for the Wachter disaster model with square-root disaster intensity.
- The formula contains the usual discounting, mean-growth, and Gaussian-risk terms plus a disaster-jump correction driven by the disaster-intensity process.
- Using Wachter's benchmark calibration, the paper finds existence for $\psi = 1.5$ and also for $\psi = 2$.

Economic message. Even though disasters are exactly the kind of tail events one might worry would explode recursive utility, the benchmark disaster calibration remains well behaved.

5.5 Proposition 5: Markov-switching learning and smooth ambiguity

Read:

- Proposition 5 reduces the Ju-Miao regime-switching learning model to a simple condition, $\ln \delta + B + \frac{1}{2}\alpha^2\sigma^2 < 0$.
- The effective growth term B lies between the good-state and bad-state growth contributions.
- Under the benchmark Ju-Miao calibration, $\Omega(1)$ is negative, so the model exists.
- The same logic implies existence for the nested Cecchetti-Lam-Mark regime-switching model.

Economic message. Once the problem is written the right way, even learning and ambiguity can yield very clean existence conditions.

5.6 Collin-Dufresne, Johannes, and Lochstoer as a failure case

Read:

- In the random-walk learning model with an unbounded conjugate normal prior, the MGF of the prior grows too fast.

- The terms in the Minkowski series therefore diverge.
- The paper proves that for $\psi \neq 1$, the model has no finite solution.
- Under smooth ambiguity aversion the divergence remains: lowering utility through ambiguity does not cure the explosive learning problem.

Economic message. This is an important negative result. The paper is not merely reassuring; it also shows that some prominent recursive-learning structures are genuinely ill defined unless the prior or the preference specification is changed.

6 Short summary

Pohl, Schmedders, and Wilms study when recursive-preferences asset-pricing models actually have a finite and unique wealth-consumption ratio. Their method combines a Minkowski bound with a Jensen comparison bound and then exploits affine structure to obtain closed-form conditions. The paper's core theoretical contribution is that existence can often be proven by checking whether a simple limit statistic, $\Omega(\cdot)$, is negative, while nonexistence can be proven when the corresponding lower bound diverges. The economically relevant solution is the finite-horizon limit and the unique no-bubble homogeneous value function. Applied to leading macro-finance models, the framework shows that standard Bansal-Yaron, Wachter, Ju-Miao, and Cecchetti-Lam-Mark calibrations are well defined, whereas a Collin-Dufresne-Johannes-Lochstoer learning model with an unbounded conjugate prior and nonunit EIS is not. The paper's broad lesson is that recursive preferences are not "plug and play": existence must be checked, but the check can often be done analytically.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the existence and uniqueness question for recursive-preferences asset-pricing models can often be answered analytically. For homogeneous recursive utilities, the relevant object is the wealth-consumption ratio. By constructing upper and lower bounds and comparing the model to a CRRA benchmark, the authors obtain practical sufficient conditions for existence and nonexistence. In the affine class, these conditions collapse to closed-form expressions that are easy to interpret and easy to evaluate.

7.2 Economic intuition

The economic intuition is that the wealth-consumption ratio is large when agents are patient, growth is strong, and long-run risks make future consumption especially valuable today. Each persistent source of risk adds to the pressure on this ratio. If that pressure becomes too strong, the infinite-horizon recursion stops being finite. The paper's bounds isolate exactly how discounting, growth, persistence, stochastic volatility, jumps, and ambiguity load into that pressure. Uniqueness then comes from insisting on the finite-horizon limit and ruling out bubble-like solutions.

7.3 Takeaway

This paper is best thought of as a preflight checklist for recursive-utility macro-finance. It tells us whether our model exists, which parameter combinations are dangerous, why some benchmark calibrations are safe, and why some seemingly innocent learning structures are not. Its lasting contribution is practical as much as theoretical: it gives asset-pricing researchers a disciplined way to verify that the model they are solving is actually there.